door to an unprecedented expansion of commerce and finance. In fact, double-entry book-keeping underpins regulation, taxation, governance, contract law, and financial regulation to this day.

Computation Over time, analytic tools saw tremendous advances at an increasing pace. One of the most significant of these advances occurred in the last century: the invention of computers. Unsurprisingly, the financial sector was among the first to adopt and use computers. For example, the IBM 650, introduced in 1954, became popular partly because of the efficiency improvements it brought in finance. In the early days of modern computing, capabilities were limited to basic arithmetic, logical and symbolic operations (for example, following "if-then" rules) to solve problems. With more computing power, analytic capabilities evolved and allowed AI to emerge from basic computer systems.

Artificial Intelligence AI broadly refers to computer systems that perform tasks that typically require human intelligence (Russell and Norvig, 2010).[1] Alan Turing and John von Neumann laid the theoretical groundwork, delineating principles that would become the cornerstone for subsequent computational and AI advancements (Turing (1950), von Neumann et al. (1945), von Neumann (1966)). For much of the 20th century, AI was dominated by GOFAI and expert systems that were developed in the wake of these seminal contributions.[2] GOFAI emerged in the late 1950s and continued to be the dominant paradigm through the 1980s. During this period, AI researchers focused on developing rule-based systems to emulate human intelligence, based on logical rules and symbolic representations. While highly useful for basic financial functions (e.g. risk management, basic algorithmic trading rules and credit scoring, fraud detection), they were far from human-level abilities in pattern recognition, handling uncertainty and complex reasoning. Hardware advances enabled small desktop computers, such as the personal computer in the 1980s and 1990s. The ability to store data and perform basic analytics using spreadsheets and other computer programs led to wide adoption and efficiency improvements in finance (Ceruzzi, 2003).

[1] The term *artificial intelligence* was first coined by the mathematician John McCarthy in a now mythical workshop at Dartmouth College in 1956.

[2] GOFAI stands for "Good Old Fashioned AI", a term coined by philosopher John Haugeland to refer to classic symbolic AI, based on the idea of encoding human knowledge and reasoning processes into a set of rules and symbols (Haugeland, 1985).

Machine Learning The next wave of progress came with machine learning (ML), a sub-field of AI (Figure 1). ML algorithms can autonomously learn and perform tasks, for example classification and prediction, without explicitly spelling out the underlying rules. Like earlier advances in information processing, ML was quick to be adopted in finance, even though in the early days, its usefulness was limited by computing power. Early examples of ML relied on large quantities of structured and labelled data.[3]

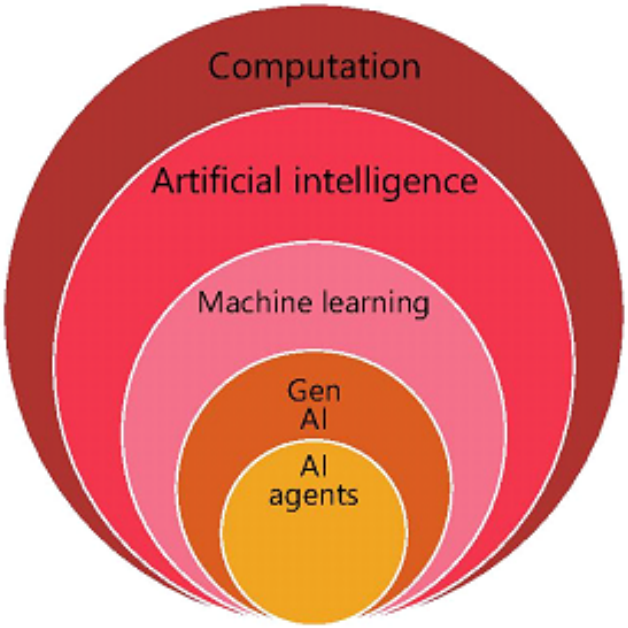


Figure 1: **Decoding AI**
Source: Authors' illustration.

The most advanced ML systems are based on deep neural networks, which are algorithms that operate in a manner inspired by the human brain.[4] Deep neural networks are universal function approximators that can learn systematic relationships in any set of training data, including increasingly in complex, unstructured

[3]Structured data refers to organised, quantitative information that is stored in relational databases and is easily searchable. It typically includes well-organised text and numeric information. Unstructured data is information that is not organised based on pre-defined models. It can include information in text and numeric formats but also audio and video. Some examples of unstructured data include text files like emails and presentations, social media content, sensor data, satellite images, digital audio, video, etc.

[4]In such systems, the input layer of artificial neurons receives information from the environment, and the output layer communicates the response; between these layers are "hidden" layers (hence the "deep" in deep learning) where most of the information processing takes place through weighted connections to previous layers.

parts of financial markets for a long time, for example, in high-frequency trading. What distinguishes the emerging generation of AI agents is that they have the intelligence and planning capabilities of cutting-edge LLMs. They can, for example, autonomously analyze data, write code to create other agents, trial-run it, update it as they see fit, and so on.[7] AI agents thus have the potential to revolutionise many different functions of financial institutions–just like autonomous trading agents have already transformed trading in financial markets.

Artificial General Intelligence For several of the leading AI labs, the ultimate goal is the development of Artificial General Intelligence (AGI), which is defined as AI systems that can essentially perform all cognitive tasks that humans can perform (Morris et al., 2024). Unlike current narrow AI systems, which are designed to perform specific tasks with a pre-defined range of abilities, AGI would be capable of reasoning, problem-solving, abstract thinking across a wide variety of domains, and transferring knowledge and skills across different fields, just like humans. Relatedly, some AI researchers and AI lab leaders speak of Transformative AI (TAI), which is defined as AI that is sufficiently capable so as to radically transform the way our economy and our society operate, for example, because they can autonomously push forward scientific progress, including AI progress, at a pace that is much faster than what humans are used to, or because they significantly speed up economic growth (Suleyman and Bhaskar, 2023). There is an active debate on whether and how fast concepts such as AGI or TAI may be reached, with strong views on both sides of the debate. As economists, we view it as prudent to give some credence to a wide range of potential scenarios for the future (Korinek, 2023b).

3 AI transforming finance

3.1 Opportunities and challenges of AI for finance

The integration of the rapidly evolving capabilities of AI is transforming the financial system. But as we have seen in Section 2, AI is just the latest information

predict without acting in the world, such as image classifiers, to have relatively low degrees of agency".

[7]Prototypes of such AI agents currently exist mainly in the realm of coding, where Devin, an autonomous coding agent developed by startup Cognition Labs, or SWE-agent, developed by researchers at Princeton (Yang et al., 2024), can autonomously take on entire software projects.